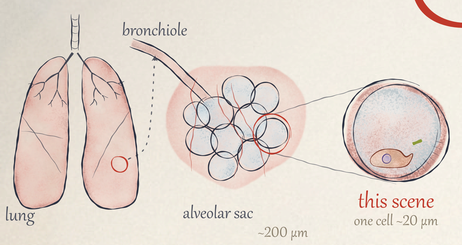
How a macrophage *destroys* what it engulfs

Cell Biology seminar · First-year Medicine
1 Aitor · 2 Nerea · 27 · 3 Urtzi · 4 Eneritz · 5 Nora · 6 Aitor

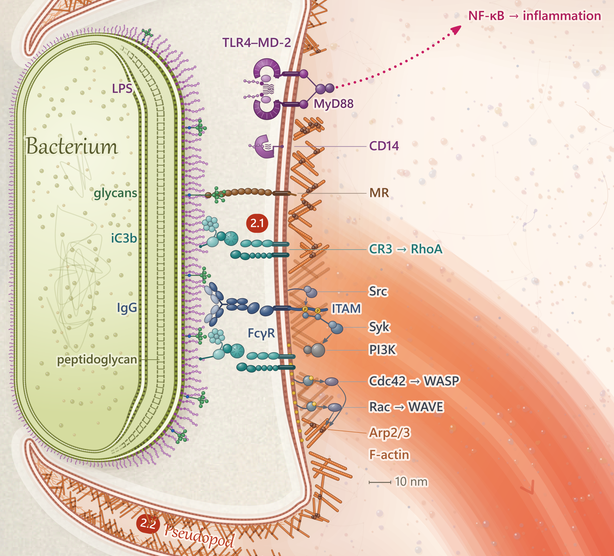
Phagocytic route · Enzyme route · Recycling
Membrane fusion · Pathological effect · Inflammatory signal
dashed = possible

Not to scale · molecular detail enlarged
cell ~20 µm · bacterium ~2 µm · receptor ~10 nm

1 Alveolar macrophage

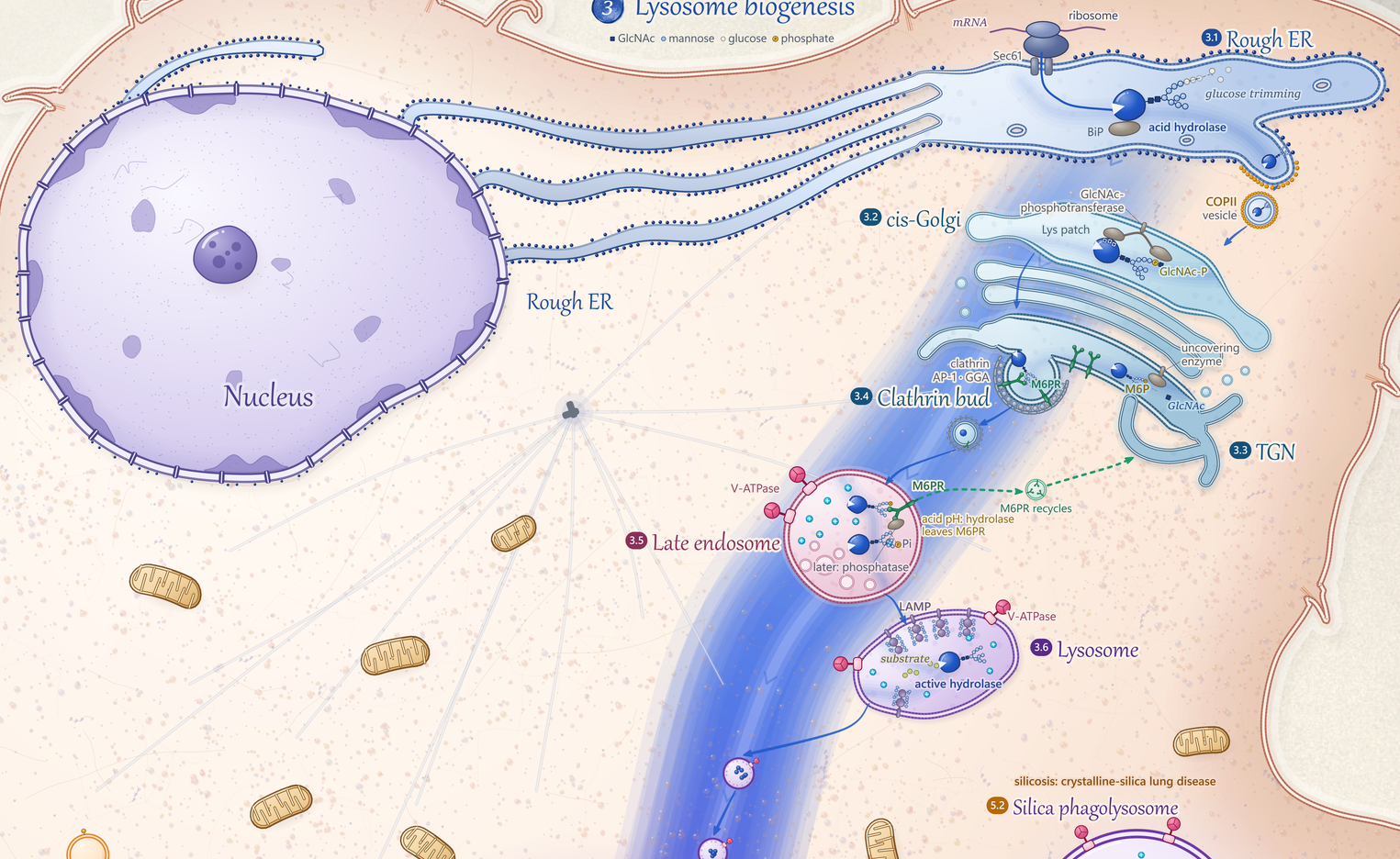


2 Recognition



3 Lysosome biogenesis

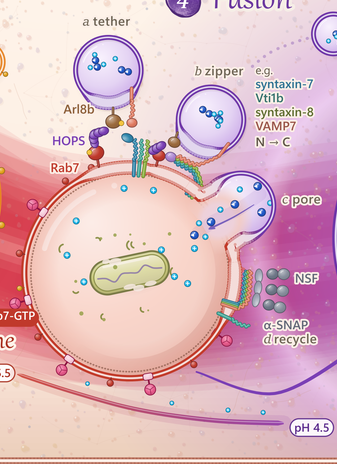
GlcNAc · mannose · glucose · phosphate



5 Particles

2.3 Nascent phagosome

4 Fusion



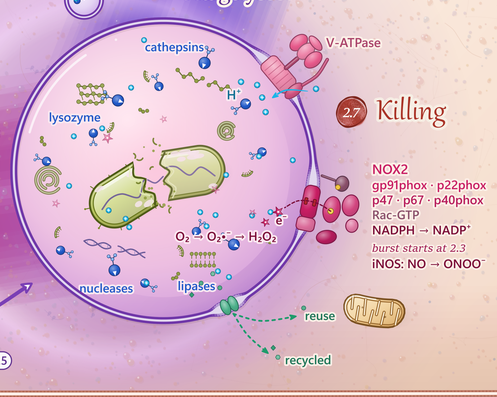
2.4 Early phagosome

pH 6.5

2.5 Late phagosome

pH 5.5

2.6 Phagolysosome



2.7 Killing

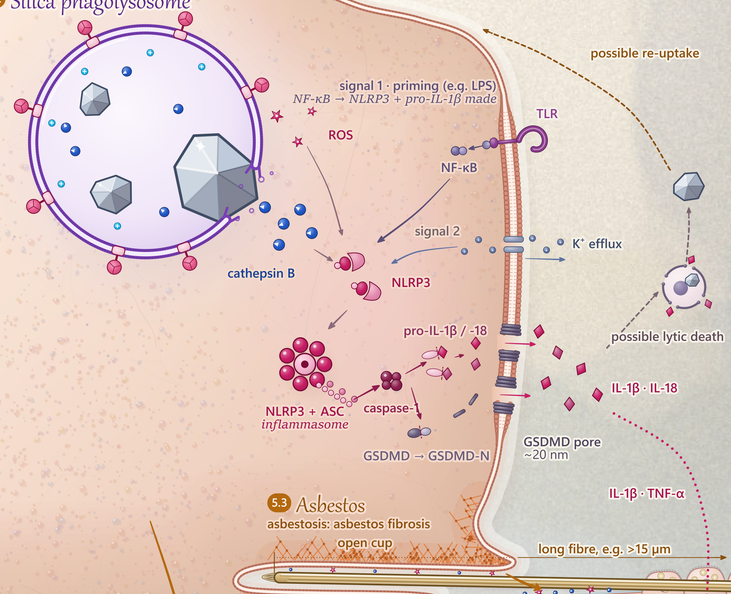
NOX2
gp91phox · p22phox
p47 · p67 · p40phox
Rac-GTP
NADPH → NADP+
burst starts at 2.3
iNOS: NO → ONOO-

pH 4.5

Mitochondrion

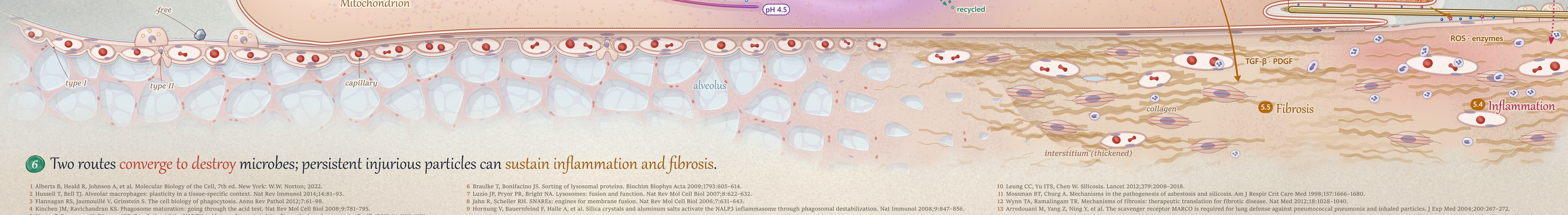
5.2 Silica phagolysosome

silicosis: crystalline-silica lung disease



5.3 Asbestos

asbestosis: asbestos fibrosis



6 Two routes converge to destroy microbes; persistent injurious particles can sustain inflammation and fibrosis.

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Successive stages in one schematic cell · not to scale; molecular detail enlarged